



CHALMERS

Task 2 Report

TME160- Multiphase Flow

Respiratory Droplets - EL



Group - EL1

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Contents

1	Introduction	2
2	STAR-CCM+ Study	3
2.1	Geometry and Computational Domain	3
2.2	Mesh Generation	4
2.3	Physical Models and Numerical Setup	4
2.4	Boundary and Initial Conditions	5
2.5	Results	6
2.5.1	Droplet dispersion during coughing	6
2.5.2	Droplet dispersion during sneezing	7
2.5.3	Effect of Ambient Air Conditions	9
2.5.4	Pathogen Footprint and Contamination Zoning	11
2.5.5	Effect of Domain Length	12
3	COMSOL Multiphysics 6.3	14
3.1	Geometry and Computational Domain	14
3.2	Mesh Generation	15
3.3	Physical Models and Numerical Setup	15
3.4	Boundary conditions	16
3.5	Results	17
4	Discussion	18
4.1	Comparison Between STAR-CCM+ and COMSOL	18
4.2	Interpretation of simulated plots	18
4.3	Limitations and future work	20
5	Conclusion	21

1 Introduction

The airborne transmission of infectious diseases occurs through the emission and transport of pathogen laden respiratory droplets generated during expiratory activities such as coughing and sneezing. Experimental and theoretical studies have shown that these violent expiratory events produce turbulent, high-momentum clouds of warm and humid air laden with droplets spanning a wide range of sizes, rather than isolated droplets following purely ballistic trajectories [1]. The dynamics of such multiphase clouds strongly influence the transport distance, residence time, and eventual fate of the droplets, thereby governing the spatial and temporal extent of contamination in indoor environments.

In this study, the dispersion of pathogen-laden respiratory droplets generated during coughing and sneezing events in an indoor environment is numerically investigated. The aim is to analyze the influence of ambient air conditions on droplet transport, spreading, and persistence, and to characterize the resulting pathogen footprint within a confined space. Previous work has shown that ambient temperature and relative humidity influence droplet evaporation and airborne persistence [2, 3], while experimental studies have also demonstrated that pathogen-carrying aerosol droplets can remain suspended in air for extended durations under typical indoor conditions [4, 5, 6].

The cough and sneeze events are modeled as transient, high momentum multiphase gas clouds emitted from the mouth, consistent with the physical picture established in prior experimental and multiphase flow studies [1]. The transport and dispersion of the liquid droplets are modeled using a Lagrangian particle tracking approach, allowing the individual droplet trajectories to be resolved within the carrier gas flow. Their characteristic properties are summarized in Table 1. Simulations are performed for three representative ambient conditions: hot and humid air (30 °C, 75% RH), cold and dry air (15 °C, 20% RH), and temperate air (20 °C, 50% RH). To assess the robustness of the numerical predictions and highlight solver-dependent differences, the study is carried out using two commercial numerical tools, STAR-CCM+ and COMSOL Multiphysics, enabling a comparative analysis of the predicted droplet dispersion behavior.

Table 1: Characteristics of cough and sneeze events used in the simulations.

Characteristic of Cough		Characteristic of Sneeze	
Temperature	34 °C	Temperature	34 °C
Relative humidity	85%	Relative humidity	95%
Density of exhaled air	0.98 kg/m ³	Density of exhaled air	0.98 kg/m ³
Density of droplets	993 kg/m ³	Density of droplets	993 kg/m ³
Velocity	11 m/s	Velocity	22 m/s
Volume fraction of droplets	1.8×10^{-7}	Volume fraction of droplets	1.8×10^{-7}
Duration	300 ms	Duration	200 ms

2 STAR-CCM+ Study

2.1 Geometry and Computational Domain

The computational domain represents a simplified indoor environment and consists of a cubic room with dimensions $2\text{ m} \times 2\text{ m} \times 2\text{ m}$. The droplet emission source is positioned on one vertical wall of the room at a height of 1.65 m from the floor, corresponding to the average mouth height of a standing adult. The overall geometry of the computational domain is shown in Figure 1. The mouth is modeled as a circular inlet with a diameter of 2 cm and is connected to a short cylindrical passage of length 5 cm . This passage ensures a well-defined inflow before the expelled jet enters the room. Droplets are injected at the outlet of this passage using a conical injector.

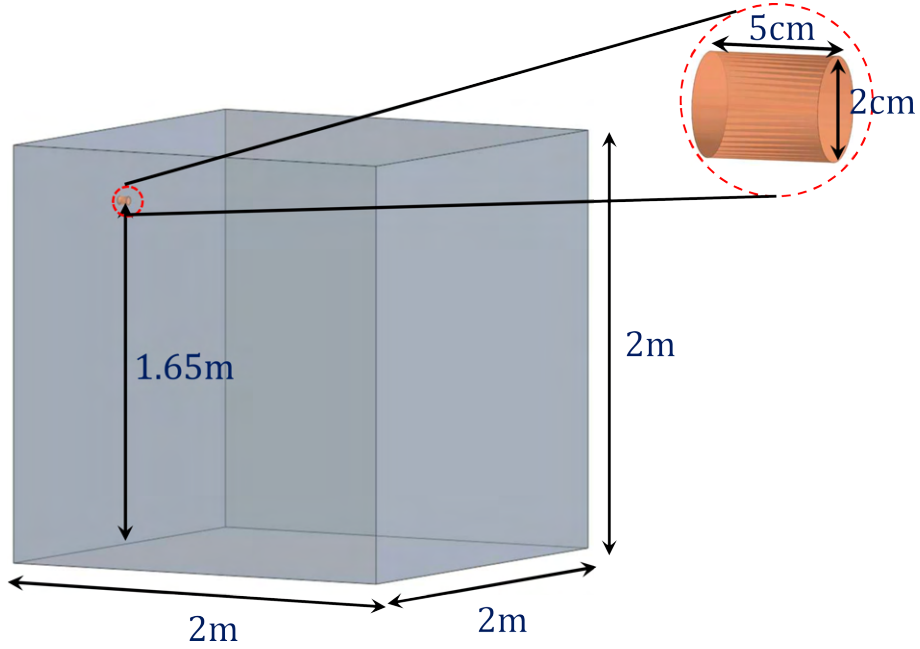
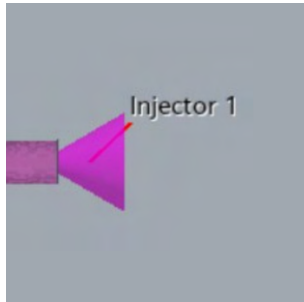
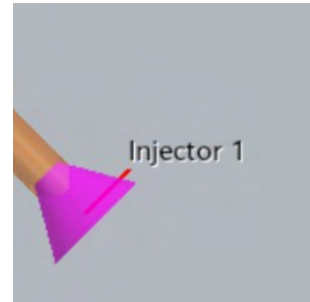


Figure 1: Computational domain.

Two injector orientations are considered to represent different expiratory events. For coughing, the injector is aligned horizontally with zero inclination, representing a forward-directed cough jet, as shown in Figure 2a. For sneezing, the injector is inclined downward at an angle of 45° as shown in Figure 2b, consistent with experimental observations indicating that sneezes are typically directed downward at angles between 40° and 50° .



(a) Horizontal injector representing coughing.



(b) Injector inclined at 45° representing sneezing.

Figure 2: Injector orientation.

2.2 Mesh Generation

The computational domain is discretized using polyhedral mesh elements. A base mesh size of 35 mm is applied throughout the room to adequately resolve the large scale flow features while maintaining reasonable computational cost. A localized mesh refinement is applied in the vicinity of the mouth passage. The refined region ensures sufficient spatial resolution near the injector, where the expelled jet and initial droplet dispersion are most sensitive to mesh resolution. A representative view of the mesh distribution, including a zoomed in view of the refined region near the mouth, is shown in Figure 3.

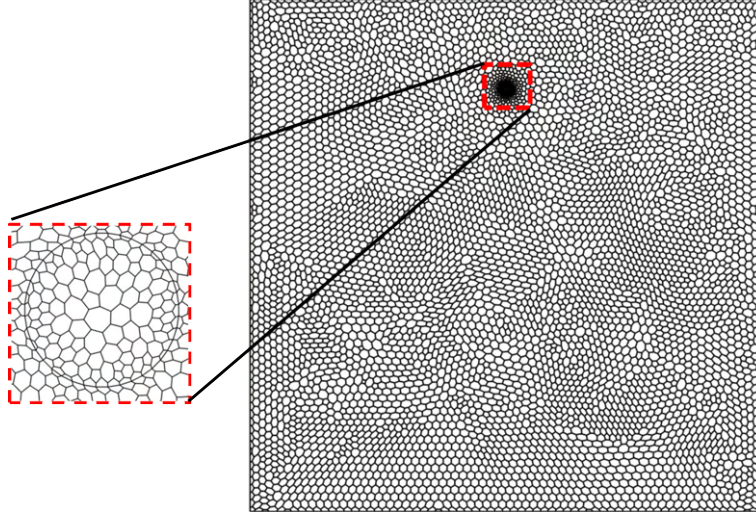


Figure 3: Discretized Domain.

2.3 Physical Models and Numerical Setup

The multiphase flow resulting from coughing and sneezing is modeled using an Eulerian Lagrangian framework. The continuous gas phase is resolved using a three-dimensional, unsteady flow model, while the dispersed liquid droplets are tracked using a Lagrangian particle approach. The selected physical models for the continuous and dispersed phases are summarized in Tables 2 and 3, respectively.

Table 2: Continuous phase modeling assumptions.

Continuous Phase Models
Three-dimensional
Unsteady-state flow
Multi-component gas
Constant density
Coupled flow solution
SST- $k\omega$ turbulence model
Eulerian formulation

Table 3: Lagrangian droplet modeling assumptions.

Lagrangian Phase Models
Constant density
liquid droplets
Drag and lift forces
TAB droplet breakup model
Two-way phase coupling
Turbulent dispersion model

The selected physical models are chosen to capture the dominant mechanisms governing respiratory droplet dispersion while maintaining numerical robustness and computational efficiency. The continuous phase is modeled as an unsteady, three-dimensional turbulent flow using an Eulerian formulation with the SST $k - \omega$ turbulence model, which is well suited for resolving high momentum expiratory jets. The dispersed phase is treated using a Lagrangian particle tracking approach, allowing size dependent droplet transport, gravitational settling, and aerodynamic interactions with the carrier flow to be resolved.

Droplet evaporation is not explicitly modeled in the present study. Instead, droplets are assumed to have constant density and fixed diameter throughout the simulations. This assumption is supported by experimental studies showing that pathogen laden aerosol droplets can persist in air for 10-30 minutes under typical indoor conditions [4], and that exhaled droplets rapidly reach their final size shortly after emission [5, 2]. Over the simulated time scales, droplet transport is therefore primarily governed by airflow dynamics and particle inertia rather than ongoing evaporation.

2.4 Boundary and Initial Conditions

The inlet boundary condition is prescribed at the entrance of the mouth passage as a transient velocity inlet. The inlet velocity, duration, and gas phase composition are specified based on the relative humidity of the exhaled air for coughing and sneezing events. The corresponding inlet conditions are summarized in Table 4.

Table 4: Inlet boundary conditions for coughing and sneezing events.

Condition	Cough (85% RH)	Sneeze (95% RH)
Gas mass fraction	H ₂ O = 0.0281, Air = 0.9719	H ₂ O = 0.0314, Air = 0.9686
Inlet velocity	11 m/s	22 m/s
Duration	300 ms	200 ms

Liquid droplets, consisting of water only, are introduced into the domain using a conical injector located at the outlet of the mouth passage. The injector parameters used for coughing and sneezing simulations are listed in Table 5.

Table 5: Injector settings for droplet injection.

Injector Parameter	Cough	Sneeze
Inner cone angle	5°	5°
Outer cone angle	60°	60°
Velocity magnitude	11 m/s	22 m/s
Volume flow rate	$6.2 \times 10^{-10} \text{ m}^3/\text{s}$	$12.4 \times 10^{-10} \text{ m}^3/\text{s}$
Injection duration	300 ms	200 ms

The initial droplet diameter distribution is prescribed using a Rosin–Rammner distribution, with particle sizes ranging from $1\mu\text{m}$ to 1 mm with reference size as $7.0\text{E-}5\text{ m}$. This distribution is used to represent the wide range of droplet sizes generated during expiratory events. The initial conditions for the continuous gas phase are prescribed based on three different ambient air conditions: hot and humid, cold and dry, and temperate air. Each case is defined by its corresponding temperature and relative humidity, which are specified in terms of water vapor and air mass fractions. In addition, a turbulent intensity of 10% and a turbulent length scale of 0.005 m are imposed uniformly throughout the domain as initial turbulence conditions. The mass fraction values used for the three ambient cases are summarized in Table 6.

Table 6: Initial ambient air conditions prescribed for the continuous phase.

Case	H ₂ O	Air
Case 1: Hot and humid air at 30 °C and 75% RH	0.0197	0.9803
Case 2: Cold and dry air at 15 °C and 20% RH	0.00213	0.99787
Case 3: Temperate air at 20 °C and 50% RH	0.00733	0.99267

2.5 Results

In this section, the numerical results obtained from the STAR-CCM+ simulations are presented. The dispersion behavior of respiratory droplets released during coughing and sneezing events is analyzed, with particular emphasis on the temporal evolution of the droplet cloud, size dependent transport mechanisms, and the resulting spatial distribution within the room.

2.5.1 Droplet dispersion during coughing

Figure 4 shows the temporal evolution of droplet dispersion following a coughing event for the hot and humid ambient condition (Case 1) at four representative time instants: $t = 0.3\text{s}$, 1s , 5s , and 9s . Droplets are displayed based on their diameter, ranging from $1\mu\text{m}$ to 1mm .

At $t = 0.3\text{s}$, a dense droplet cloud is observed close to the mouth opening, with droplets of all sizes transported primarily along the cough jet. By $t = 1\text{s}$, differences in the trajectories of droplets with different diameters become visible. Larger droplets begin to move downward, while smaller droplets remain aligned with the main flow direction. At later times ($t = 5\text{s}$ and 9s), a clear spatial separation is observed. Larger droplets are predominantly located in the lower regions of the domain, whereas smaller droplets remain suspended and spread over a wider area of the room. The results show a clear dependence of droplet trajectories on particle

diameter, with larger droplets predominantly located in the lower regions of the domain at later times, while smaller droplets remain distributed throughout the room.

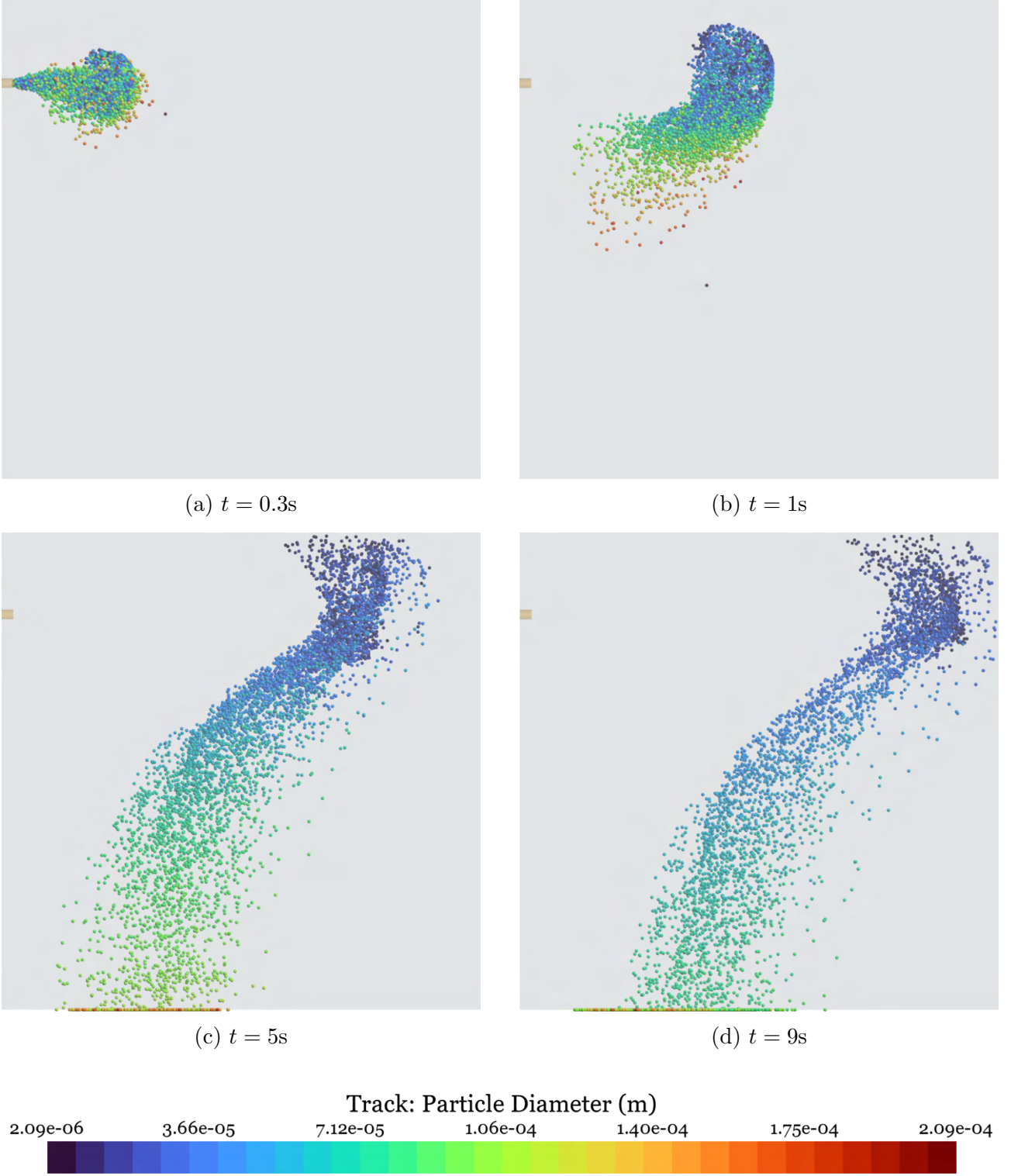


Figure 4: Time evolution of droplet dispersion during a coughing event. Droplets are colored by particle diameter.

2.5.2 Droplet dispersion during sneezing

Figure 5 shows the temporal evolution of droplet dispersion following a sneezing event for the hot and humid ambient condition (Case 1) at four representative time instants: $t = 0.2s$, $1s$,

3s, and 5s. Droplets are displayed based on their diameter, ranging from $1\mu\text{m}$ to 1mm .

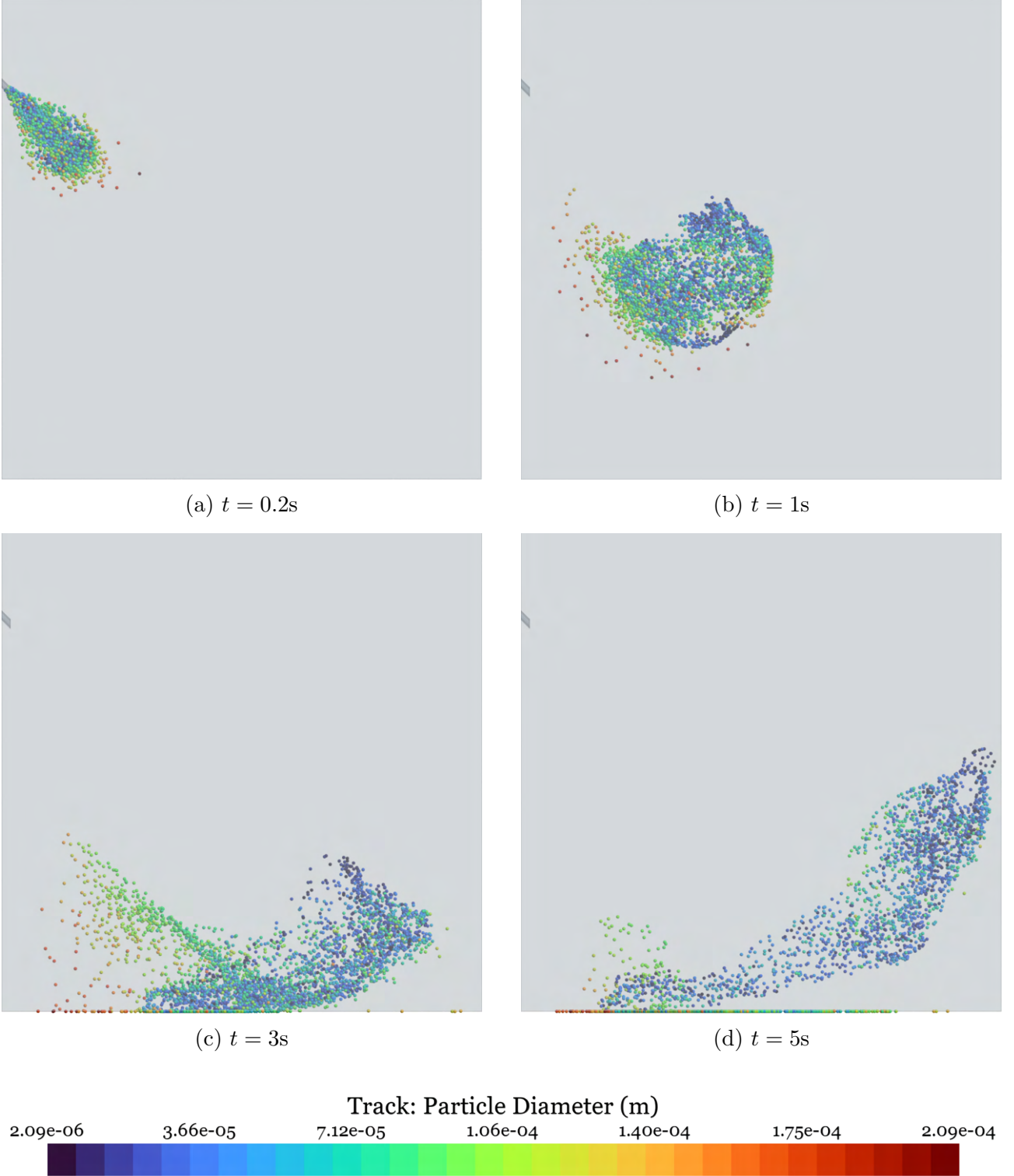


Figure 5: Time evolution of droplet dispersion during a sneezing event.

At $t = 0.2\text{s}$, a highly concentrated droplet cloud is observed close to the mouth opening, extending rapidly into the room. By $t = 1\text{s}$, size-dependent differences in droplet trajectories become visible. Larger droplets begin to deviate downward from the main flow direction, while smaller droplets remain aligned with the jet. At later times ($t = 3\text{s}$ and 5s), the droplet cloud occupies a larger spatial region compared to the coughing case. Larger droplets are

predominantly located in the lower regions of the domain, whereas smaller droplets remain suspended and distributed over a wider volume.

The penetration distance and spatial extent of the droplet cloud are greater for the sneezing event, consistent with the higher injection velocity of 22m/s compared to coughing.

2.5.3 Effect of Ambient Air Conditions

For cough: Figure 6 presents the particle streamlines representing droplet dispersion during a coughing event over a duration of 10s for three different ambient air conditions: hot and humid air (Case 1), cold and dry air (Case 2), and temperate air (Case 3). Droplets are displayed based on their diameter, ranging from $1\mu\text{m}$ to 1mm.

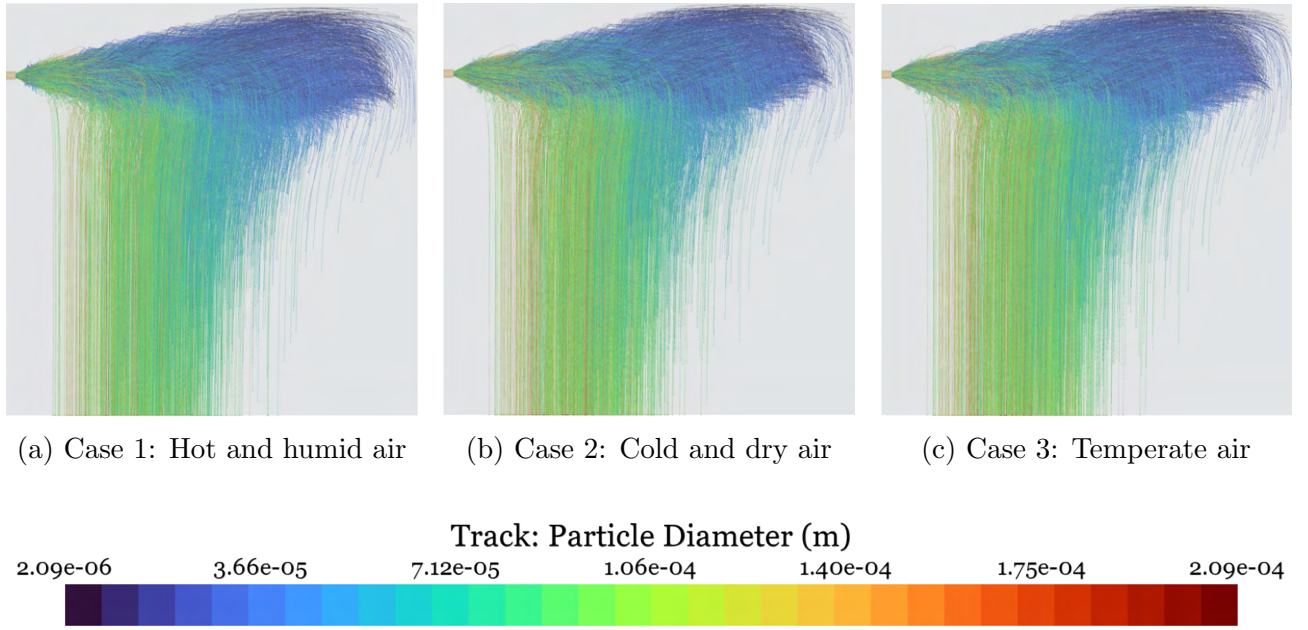


Figure 6: Effect of ambient air conditions on droplet dispersion during a coughing event.

For the hot and humid condition (Case 1), the particle streamlines show a relatively compact droplet cloud close to the source. Streamlines corresponding to larger droplets extend downward and reach the lower regions of the domain, while smaller droplets remain suspended and are transported further into the room. In the cold and dry condition (Case 2), a similar size-dependent dispersion pattern is observed. However, streamlines corresponding to larger droplets reach the ground closer to the source compared to Case 1. Larger droplets are more concentrated near the lower region of the domain, while smaller droplets remain distributed within the airflow. For the temperate condition (Case 3), the particle streamline pattern is largely similar to that observed in Case 2. Larger droplets are predominantly located near the lower regions of the domain, whereas smaller droplets remain airborne and dispersed throughout the room. Minor differences in the spatial distribution of droplets are observed compared to the other cases.

For sneeze: Figure 7 presents the particle streamlines representing droplet dispersion during a sneezing event over a duration of 5s for three different ambient air conditions: hot and humid air (Case 1), cold and dry air (Case2), and temperate air (Case 3). Droplets are displayed based on their diameter, ranging from $1\mu\text{m}$ to 1mm.

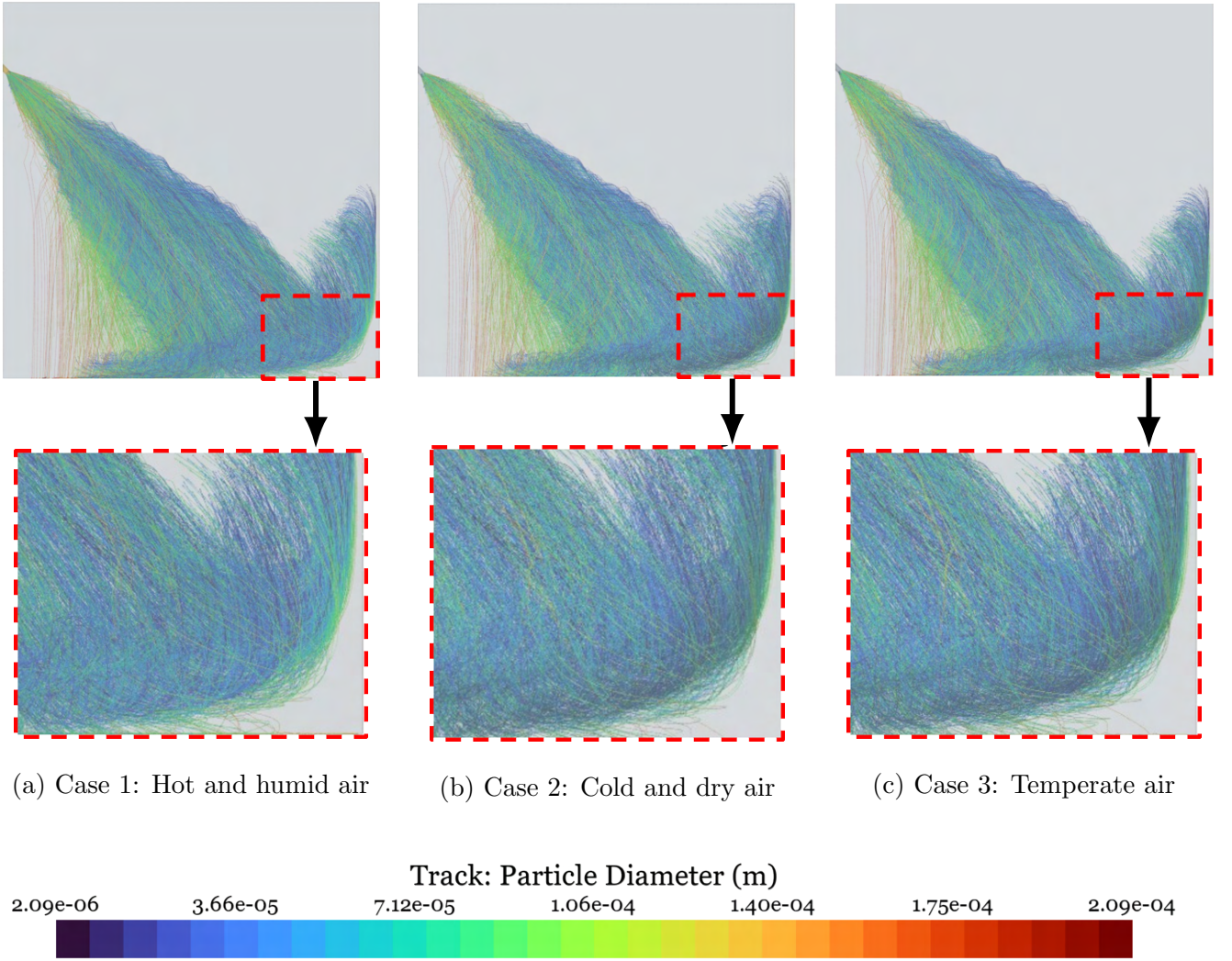


Figure 7: Effect of ambient air conditions on particle streamlines during a sneezing event. The arrows indicate the regions enlarged in the corresponding zoomed views.

The particle streamline patterns obtained under the three ambient air conditions show similar large scale dispersion behavior. In all cases, the droplet cloud extends deep into the room, occupying a comparable spatial region across the three ambient conditions. A closer inspection of the highlighted downstream region, shown in the zoomed views, reveals minor differences between the cases. For the hot and humid condition (Case 1), streamlines corresponding to larger droplet diameters are observed further downstream, indicating that larger droplets are present at greater distances from the source. In contrast, for the cold and dry (Case 2) and temperate (Case 3) conditions, the highlighted region is dominated primarily by smaller droplets. Streamlines corresponding to larger droplets are less prominent, and larger droplets are predominantly located closer to the source or near the lower regions of the domain.

2.5.4 Pathogen Footprint and Contamination Zoning

For Cough: Figure 10 shows the droplet distribution resulting from a coughing event over a duration of 10s. The droplet cloud spreads forward from the source and occupies a limited region of the room, with particles distributed throughout the flow domain. A noticeable accumulation of droplets is observed near the lower region of the room, indicating particle settling during the simulation period. The corresponding ground contamination pattern reveals that droplet deposition is primarily confined to a localized area downstream of the source. The highest concentration of deposited particles is observed close to the coughing individual, with deposition density decreasing progressively with distance along the floor.

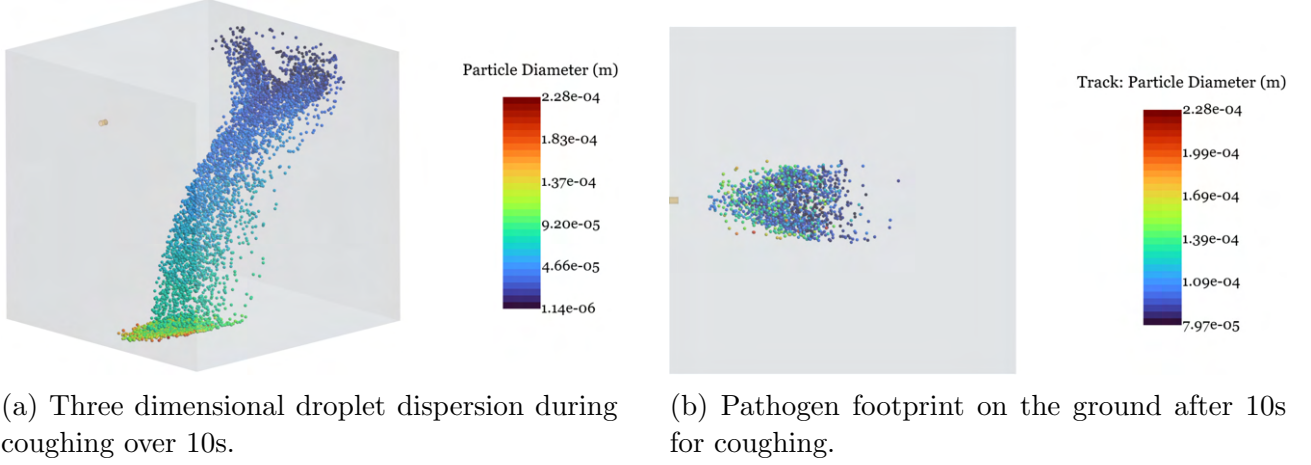


Figure 8: Pathogen footprint and contamination zoning resulting from a coughing event over a duration of 10 s.

For Sneeze: Figure 9 presents the ground contamination pattern produced by a sneezing event over the same duration of 5s. Deposited droplets are distributed over a larger area of the floor compared to the coughing case. The contamination region extends further downstream from the source, indicating a broader spatial spread of deposited particles at ground level.

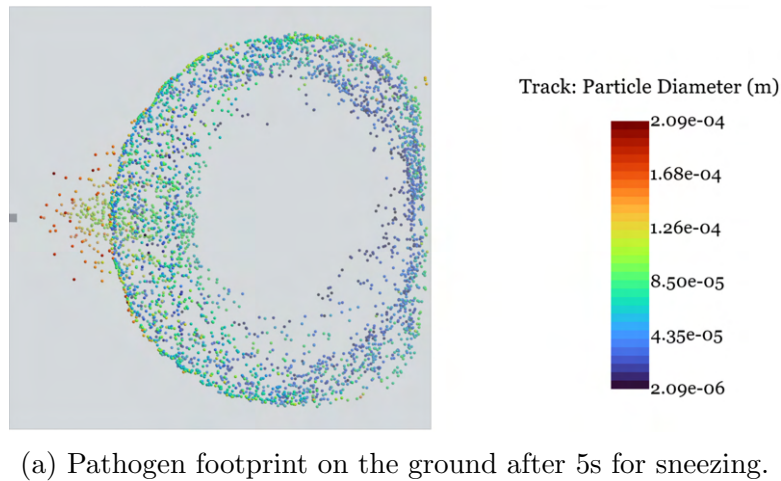


Figure 9: Pathogen footprint and contamination zoning resulting from a sneezing event over a duration of 5s.

2.5.5 Effect of Domain Length

The length of the computational domain plays an important role in accurately predicting the dispersion and deposition of respiratory droplets generated during coughing and sneezing events. In shorter computational domains, the outlet boundary is located closer to the emission source which may lead to premature truncation of the droplet plume and an under prediction of downstream transport.

The reference diameter of the particles were reduced to $4\text{e-}5\text{ m}$ to increase the number of small particles in the spray and so that their behaviour can be monitored in this larger domain. Figure 10 illustrates the effect of domain length on the spatial distribution of droplets for coughing and sneezing cases.

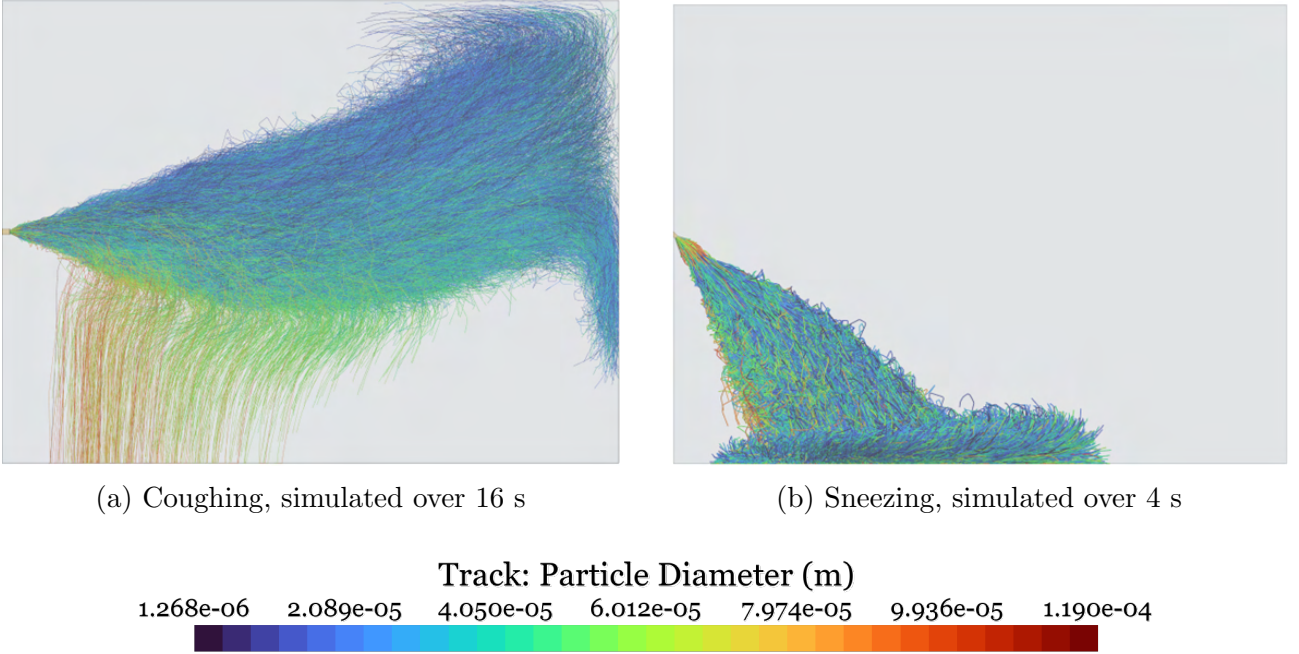


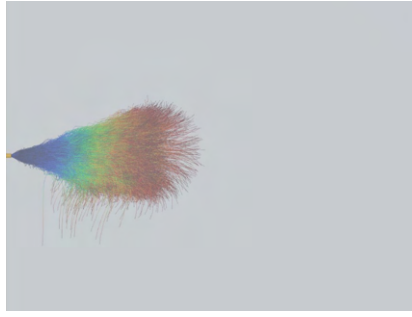
Figure 10: Results for 4 m domain length

As the domain length is increased, the airflow field and droplet trajectories are allowed to develop more naturally enabling better capture of long range transport and settling behavior. This results in a larger spatial dispersion of droplets and an extended pathogen footprint on the ground particularly for smaller droplets that remain suspended in the air for longer durations. Further increases in the domain length beyond a certain value produce only marginal changes in the predicted droplet dispersion and deposition patterns, indicating that the solution becomes independent of the domain size.

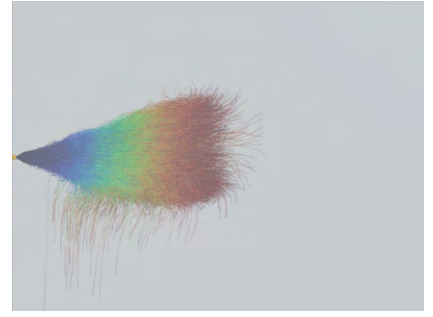
Figure 11 Shows the particle trajectories at different time steps and it is clearly evident that the smaller particles are carried further by the induced turbulence in the ambient air as the larger particles leave the core flow. This effect was investigated and photographed in [1]



(a) $t = 0.3$ s



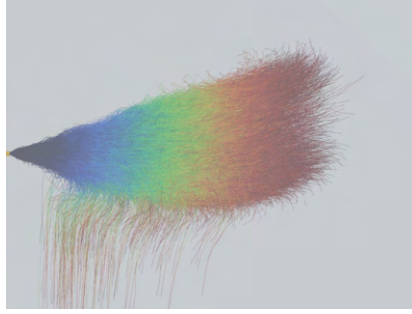
(b) $t = 1$ s



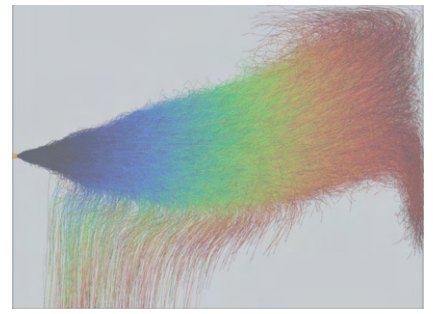
(c) $t = 3$ s



(d) $t = 5$ s



(e) $t = 9$ s



(f) $t = 16.2$ s

Figure 11: Particle tracks at different time steps.

3 COMSOL Multiphysics 6.3

3.1 Geometry and Computational Domain

COMSOL Multiphysics is a program that is based of finite element method, which divides a continuous medium into small elements. The program is usually used for solving problems that has even and continuous fields such as even temperature distribution or laminar flows. However, in this project the program was used to simulate a turbulent flow with particles that has stochastic effects. The computational domain was designed to represent a simplified indoor environment, consisting an inlet passage. A cough was simulated from the inlet passage, consisting an injector with a shape of a cone.

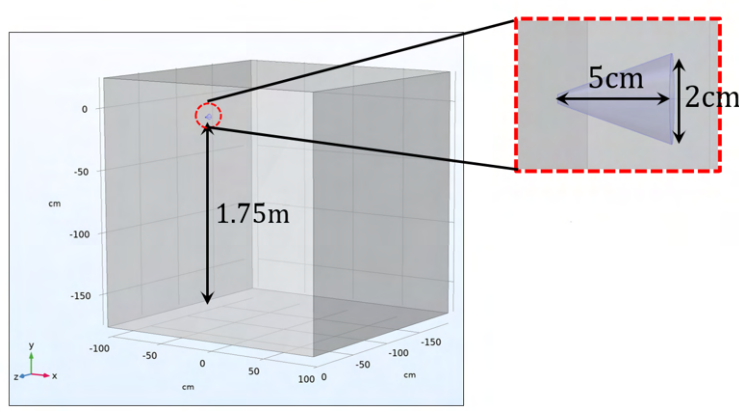


Figure 12: Geometry of the simplified indoor environment

Droplets were injected from a point source located at the inlet, mimicking a cough with an exhaled air temperature of 34°C and a inlet velocity of 11 m/s . The exhaled air is a stationary turbulent field which was assumed for the carrier phase. Ambient conditions varied across three cases according to the task description. The exhaled air properties remained constant throughout the three different cases. The contained exhaled air with an outlet at the end of the cone. The room contained the ambient air, which was changed depending on the case that was studied.

The governing forces that were included in the study was drag, lift, gravity and buoyancy. All forces were set to affect all particles. The gravity and buoyancy force are included in the same equation. ρ_p is the density of the particle and ρ is the density of air. The equation is following:

$$\mathbf{F}_g = m_p \mathbf{g} \frac{\rho_p - \rho}{\rho_p} \quad (1)$$

The drag force was defined as:

$$\mathbf{F}_D = \frac{1}{\tau_p} m_p (\mathbf{u} - \mathbf{v}) \quad (2)$$

$$\tau_p = \frac{\rho_p d_p^2}{18 \mu} \quad (3)$$

The lift force was defined as:

$$\mathbf{F}_L = 6.46 r_p^2 \mathbf{L}_v \sqrt{\mu \rho \frac{|\mathbf{u} - \mathbf{v}|}{|\mathbf{L}_v|}} \quad (4)$$

$$\mathbf{L}_v = (\mathbf{u} - \mathbf{v}) \times [\nabla \times (\mathbf{u} - \mathbf{v})] \quad (5)$$

However, the hydropscopic growth and evaporation were neglected since it was not possible option to examine in COMSOL. This limitation may have had an effect in that the particle size remained almost constant in the three different cases, regardless of the ambient humidity.

3.2 Mesh Generation

The room was set to have a 'no slip' wall condition. The wall condition of the cone was set to 'bounce'. The meshing of the entire geometry was set to a user controlled mesh. The cone had a maximum element size of 0,05 cm and minimum 0,01 cm.

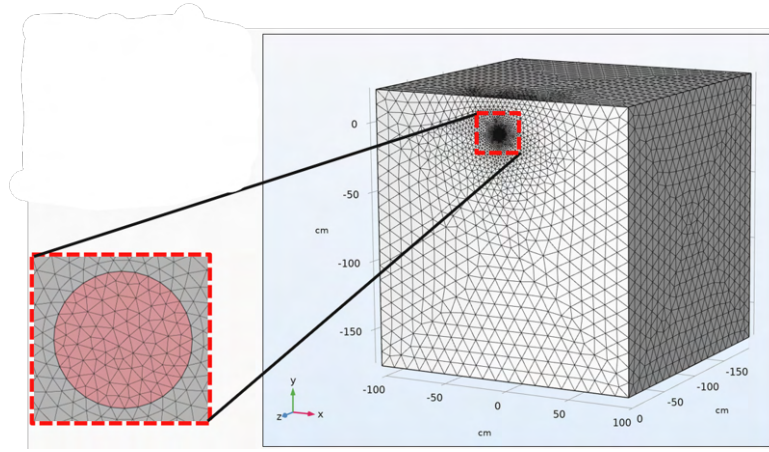


Figure 13: Meshing of the geometry

3.3 Physical Models and Numerical Setup

The flow of the cough have been generated by using Eulerian-Lagrangian framework as a model. The carrier phase is resolved through a three-dimensional, steady-state turbulent flow model. A summary of the models can be seen in the tables 7.

Table 7: Continuous phase modeling assumptions.

Continuous Phase Models
Three-dimensional
Steady-state flow
Multi-component gas
Constant density
Coupled flow solution
Algebraic yPlus turbulence model
Eulerian formulation

A particle tracing study was made to simulate the droplets released from the injector, as can be seen in table 8. The liquid droplets are tracked by using Lagrangian Particle tracking. The particle distribution followed a normal distribution of $28\text{ }\mu\text{m}$ and a standard deviation of $10.3\text{ }\mu\text{m}$, since the Rossin-Ramler distribution was not an available option in COMSOL. All particles were assigned the same density 993 kg/m^3 . Wall interactions were modeled using a bounce condition, with a default setting of 10 000 secondary particles. This means that particles that hit a wall after 10 000 times will be deleted by the program in order for COMSOL to be able to simulate by making these simplifications and limitations. Otherwise the simulations would take a long time to compute.

Table 8: Lagrangian droplet modeling assumptions.

Lagrangian Phase Models
Constant density
Time-dependent flow
liquid droplets
Drag, lift, buoyancy and gravity force
Two-way phase coupling
Turbulent dispersion model

3.4 Boundary conditions

The boundary conditions have been summarized in table 9. It is important to note that only two cases were studied. This is because COMSOL kept crashing and we were not able to make accurate simulations.

Table 9: Initial ambient air conditions prescribed for the continuous phase.

Case	Water content Air breath (kg/m^3)	Water content Ambient Air (kg/m^3)
Case 1: Hot and humid air at $30\text{ }^\circ\text{C}$ and 75% RH	0.0298	0.0225
Case 2: Cold and dry air at $15\text{ }^\circ\text{C}$ and 20% RH	0.0298	0.006

Table 10 summarizes the settings that were made for the cone. The program was unable to have a injector durations longer than 100 ms, due to program crashing.

Table 10: Injector settings for droplet injection.

Injector Parameter	Cough
Cone angle	20°
Velocity magnitude	11 m/s
Injection duration	100 ms

3.5 Results

Figure 14 presents particle trajectories for two coughing scenarios simulated in COMSOL under different ambient conditions. The color scale indicates particle diameter in centimeters, warmer colors represents larger droplets and vice versa.

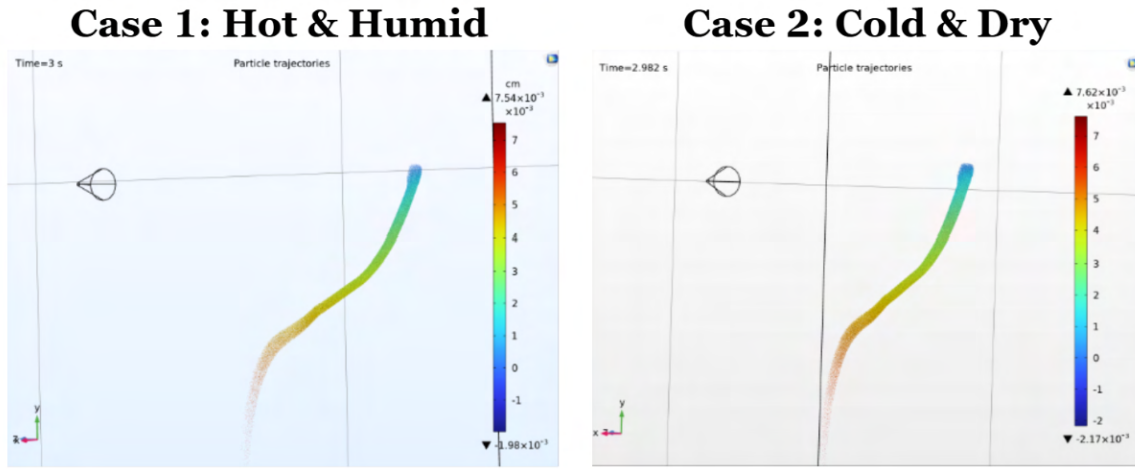


Figure 14: Results for a cough in COMSOL. The particle diameter is plotted in the graph with a corresponding legend showing particle size in cm.

The two images appear different due to different viewing angles, but they have the same particle distributions. In both scenarios, larger droplets remain close to the source, as shown by the color gradient along the trajectories.

4 Discussion

4.1 Comparison Between STAR-CCM+ and COMSOL

Since COMSOL the FEM-method, it is less suitable for problems involving discrete and highly transient flows. Particles are discrete objects and not a continuous medium. Particles can have different velocities, directions and interactions. The FEM-method has therefore a hard time describing these stochastic effects. Because of this, COMSOL failed to capture realistic particle dispersion patterns. This can be seen in the simulations, where the particles move in only one direction and spread throughout the room with a too slow speed. The conclusion can be made that COMSOL is more relevant to use for studies that have steady-state or low turbulent flows.

COMSOL is built upon the FEM method to capture the fluxes using element wise distribution of the control volume, which has weak turbulence-particle coupling. The cough/sneeze phenomena are heavily dominated by highly unsteady turbulent jets with strong velocity fluctuations in COMSOL. The particle tracking typically uses mean flow fields, and turbulent dispersion is added manually, resulting in particles following overly smooth trajectories and under-disperse. FEM time stepping tends to smooth sharp gradients, particle tracking loosely coupled under the flow resulting in peak momentum and jet penetration underestimated.

FEM meshes are too coarse in shear layers, they introduce numerical diffusion the jet is too fast in the simulation, hence the particles fall out unrealistically. The drag models are limited in COMSOL, the evaporation and humidity are harder to implement. FEM schemes introduce numerical smoothing.

However, Star CCM+ uses Finite Volume Method, which is well suited for transient, turbulent and multiphase flows. Both Star CCM+ and COMSOL use Lagrangian Particle tracking, but Star CCM+ can compute stochastic effects more accurately than COMSOL. The reason Star CCM+ can capture stochastic effects more accurately is because of the software's numerical approach and turbulence modeling capabilities. The FVM method is more preferable than the FEM method, since it solves equations over discrete control volumes. This is preferable for transient flows and gives a more accurate resolution of turbulence through models. These turbulence models introduce fluctuating velocity components that interact with particles, creating realistic dispersion patterns.

Star CCM+ uses turbulence dispersion models in the Lagrangian Particle tracking, where the simulations will take account for turbulent eddies and their effect on particle trajectories. The particles will therefore spread in different directions. However, since COMSOL lacks turbulence driven randomness in the Lagrangian framework, the results of the simulation will therefore show trajectories in the same direction and a unrealistic dispersion of particles.

4.2 Interpretation of simulated plots

Figures 4, 6, and 14 show a consistent droplet dispersion behavior during a coughing event across both the STAR-CCM+ and COMSOL simulations. In all cases, a clear size dependent separation of droplets is observed. Larger droplets tend to deviate downward from the main jet and settle toward the lower regions of the domain, while smaller droplets remain suspended and are transported with the airflow generated by the cough.

This behavior is governed by the balance between gravitational and aerodynamic forces acting on the droplets. Since all droplets are modeled with the same material density, the separation of trajectories is driven by particle diameter and volume. Larger droplets experience a

significantly stronger gravitational force, which scales with particle volume, causing them to overcome drag more rapidly and settle earlier. In contrast, smaller droplets experience a much lower gravitational force relative to drag and buoyancy, allowing them to remain coupled to the carrier phase flow and follow the turbulent jet structure for longer durations. The observed size dependent response can also be interpreted using the Stokes number. Larger droplets have a higher particle response time, resulting in a larger Stokes number, and therefore tend to decouple from the flow and follow more ballistic trajectories. Smaller droplets, which are characterized by a lower Stokes number, respond more quickly to variations in the surrounding airflow and remain closely coupled to the continuous phase. This explains why smaller droplets spread over a larger volume of the room, while larger droplets remain confined closer to the source and the ground.

For sneezing events, a similar size dependent dispersion behavior is observed, as shown in Figures 5 and 7. Due to the higher injection velocity associated with sneezing, the droplet cloud penetrates further into the domain and occupies a larger spatial region compared to coughing. Despite this increased momentum, the fundamental dispersion mechanisms remain unchanged: larger droplets settle rapidly under gravity, while smaller droplets remain airborne and are transported by the turbulent flow.

When comparing the three ambient air conditions : hot and humid, cold and dry, and temperate the particle trajectories shown in Figures 6 and 7 exhibit similar large scale dispersion patterns for both coughing and sneezing events. In all cases, the overall shape and extent of the droplet cloud are primarily governed by the initial jet momentum, gravitational settling, and aerodynamic drag. As a result, the global dispersion behavior does not change dramatically between the ambient conditions. However, a closer inspection of the particle streamlines reveals subtle but consistent differences, particularly for larger droplets. For the coughing event shown in Figure 6, larger droplets in the hot and humid case tend to remain entrained in the flow for a longer distance before settling, leading to deposition further downstream compared to the cold and dry case. In contrast, under cold and dry conditions, larger droplets settle more rapidly and reach the ground closer to the source. The temperate case generally exhibits behavior that lies between these two conditions.

The interpretation of the simulated plots indicates that the droplet size and injection momentum are the dominant factors governing respiratory droplet dispersion in the study. Variations in ambient air condition have a comparatively limited influence under the adopted modelling assumptions. The observed difference between coughing and sneezing are primarily driven by difference in initial jet velocity.

This size dependent dispersion behavior is directly reflected in the resulting pathogen footprint and contamination zones shown in Figures 10 and 9. Larger droplets, which settle rapidly under gravity, tend to deposit close to the source, leading to a region of higher contamination in the near field. In contrast, smaller droplets remain airborne for longer durations and are transported over a larger portion of the room, resulting in a more widespread but lower-intensity contamination pattern. For sneezing events, the higher initial jet velocity extends the contamination footprint further downstream compared to coughing, while preserving the same near-field dominance of larger droplets and far-field transport of smaller droplets.

4.3 Limitations and future work

Limitations

Despite providing useful insights into the behavior of respiratory droplets, this study is subject to several limitations related to modeling assumptions, numerical methods, and computational constraints.

- **Physical and Modeling Assumptions:** The simulations assume simplified droplet properties, including a uniform size distribution and spherical droplet shapes. Evaporation effects were neglected, which may influence droplet lifetime and dispersion, especially for smaller droplets.
- **Numerical and Software Limitations:** Differences in numerical solvers, mesh generation techniques, and multiphase models between STAR-CCM+ and COMSOL Multiphysics may lead to discrepancies in the results. Mesh resolution was limited by computational resources and particle size selection, potentially affecting the accuracy of the model.
- **Validation Constraints:** Due to the limited availability of experimental data under identical conditions, full quantitative validation of the numerical results was not possible. The findings should therefore be interpreted as qualitative and comparative rather than absolute predictions.

Future Work

Several improvements and extensions can be considered to enhance the accuracy and applicability of the present study.

- **Advanced Droplet Physics:** Future studies could incorporate droplet evaporation, heat and mass transfer, and non-spherical droplet behavior. Including breakup and coalescence models would provide a more realistic representation of respiratory aerosols.
- **Environmental Effects:** The influence of ambient conditions such as humidity, temperature, and ventilation flow patterns should be studied in detail. This would allow better assessment of droplet dispersion in indoor environments.
- **Validation and Mesh Refinement:** Future work should focus on validation against experimental or clinical data. Performing mesh independence studies and using higher-resolution meshes would improve the accuracy of numerical predictions.
- **Extended Comparative Studies:** Further comparisons between different CFD solvers and multiphase models could help identify the strengths and limitations of each software package, supporting more informed tool selection for respiratory flow simulations.

5 Conclusion

In this study, the dispersion of pathogen laden respiratory droplets generated during coughing and sneezing events in an indoor environment was numerically investigated using STAR-CCM+ and COMSOL Multiphysics. The simulations focused on analyzing the effects of droplet size, injection momentum and ambient air conditions on droplet transport, spreading and deposition.

The results show a clear size dependent dispersion behavior for both coughing and sneezing events. Larger droplets deviate from the main jet and settle rapidly under the influence of gravity, leading to localized deposition near the source. In contrast, smaller droplets remain airborne for longer durations and are transported over a larger volume of the room, contributing to widespread but lower intensity contamination. Sneezing events produce a more extensive droplet cloud and a broader contamination footprint compared to coughing, primarily due to the higher initial jet velocity.

Variations in ambient air conditions (hot and humid, cold and dry, and temperate) were found to have a limited influence on the overall droplet dispersion patterns. While minor differences were observed in the settling behavior of larger droplets, the global transport of the droplet cloud was mainly governed by the initial jet momentum, gravitational settling and aerodynamic drag. Increasing the domain length facilitated tracking of the behaviour of smaller particles which are carried further by the ambient air which explains how the pathogen containing particles can contaminate and travel large distances. This requires more computational power and time

The pathogen footprint analysis further highlights that near field contamination is dominated by larger droplets depositing close to the source, whereas far field exposure is primarily associated with smaller droplets that remain suspended and disperse throughout the room. Overall, the study demonstrates that droplet size and injection momentum are the primary factors controlling respiratory droplet dispersion under the present modeling assumptions providing qualitative insight into contamination patterns in indoor environments.

As previously mentioned both Star CCM+ and COMSOL were used for the calculations. The two different software packages produced different results and it is notable that Star CCM+ is a more suitable software for simulation of high intensity events such as the cases studied. The main differences between COMSOL and Star CCM+ are that COMSOL is FEM based while Star CCM+ uses the FVM. COMSOL is also much more limited in terms of available models and takes more of a "black box" approach than Star CCM+. Although the results from COMSOL were worse, some general trends, such as the strong size dependency of the particles are still visible in the results.

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